

CHANDRIL MALICK

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SUMMARY

AI product engineer who has designed and shipped full-stack, production-grade AI systems end to end — from a privacy-first RAG tutoring platform running fully offline on local LLM inference, to a multimodal diagnostic pipeline combining computer vision and tabular ML for real-time health monitoring. Comfortable across the full stack: retrieval architecture, model training, and FastAPI-based deployment.

EXPERIENCE

FlyRank AI — Machine Learning Engineering Intern (Jul 2026 – Aug 2026)

- Engineered semantic retrieval and intent-classification pipelines across **50,000+ search queries** and **2 GB+ of search logs**, using Python, Sentence-Transformers, and Scikit-learn to evaluate and improve search relevance.
- Built embedding-based evaluation pipelines that cut manual relevance-review time from **4 hours to 45 minutes per evaluation cycle**, enabling analysis across **20+ production search workloads**.

Samsung Innovation Campus — AI Intern (Sep 2025 – Nov 2025)

- Architected end-to-end PyTorch and TensorFlow NLP pipelines covering preprocessing, feature engineering, and supervised training across **25,000+ labeled records**, achieving **90%+ evaluation accuracy**.
- Deployed trained models through FastAPI REST APIs, reducing inference latency from **800 ms to 520 ms** while supporting **1,000+ API requests/day**.

PROJECTS

DABBA AI — Privacy-First Educational AI [Live](#) | [GitHub](#)

- Designed and deployed a Retrieval-Augmented Generation (RAG) tutoring platform, initially **testing at Brainware University** for real-world student use, with semantic retrieval across **10,000+ educational documents** using LangChain and FAISS.
- Integrated local **Gemma 3:1B** inference to enable privacy-preserving, offline-capable tutoring without relying on cloud-based LLM inference.

SmartSant-IoT — Multimodal Disease Prediction [Live](#) | [GitHub](#)

- Built a multimodal diagnostic platform combining medical imaging and clinical data, training and evaluating EfficientNet-B0 and LightGBM models to achieve 93.06% accuracy on urinary tract infection detection.
- Shipped scalable preprocessing and FastAPI microservice pipelines with SMOTE-based class balancing to support real-time health monitoring.

TECHNICAL SKILLS

Languages: Python, C

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy

LLMs & GenAI: RAG, Prompt Engineering, LangChain, Hugging Face, FAISS, Semantic Search, Embeddings, Vector Databases

Backend: FastAPI, REST APIs

Frontend: React.js, HTML, CSS, Tailwind CSS

Databases: MySQL, SQLite, Firebase

Cloud & DevOps: AWS (EC2), Docker, Git, GitHub, Linux, CI

PUBLICATION

IEEE Conference Publication (2026): "A Low-Cost Smart IoT-Enabled Multimodal Sanitation System for Health Monitoring and Urinary-GI Disorders Detection via Bio-Waste Analysis."

[View on IEEE Xplore](#)

ACHIEVEMENTS

- Top 1,000 Innovators** among **10,000+** nationwide participants, SBI Youth Ideathon 2025 — for SmartSant-IoT.
- Top 25 Finalist**, HP Dreams Unlocked (Tech Track 2025) — for DABBA AI.
- Finalist, Brainware AI-Hackathon; Participant, Hack4Delhi, CONVOLVE 4.0, and Code Clash.

CERTIFICATES & COURSES

- Web Development Course — Apna College | Data Structures & Algorithms — Apna College | Python Course — Kaggle
- IBM Industrial Training — IBM (Jul 2026 – Aug 2026) | Microsoft Certified Professional — Cloud Completion

EDUCATION

Brainware University, Kolkata, India — B.Tech, Computer Science & Engineering (Aug 2023 – Jul 2027)

CGPA: 8.3/10 | Coursework: DSA, Algorithms, DBMS, AI, Operating Systems